Série_5

Exercice 1

1) Pb : A est décomposable en $LL^T = A$

Défini positif $\Leftrightarrow$ dét $A_k > 0$ $k = 1 \dots 4$

$q(x) = X^T A x > 0$, $x \in \mathbb{R}^4 - \{0_4\}$

$A = L . L^T$

$$L = \begin{pmatrix} l_{11} & & & 0 \\ l_{21} & l_{22} & & \\ 0 & l_{32} & l_{33} & \\ 0 & 0 & l_{43} & l_{44} \end{pmatrix}$$

$$L^T = \begin{pmatrix} l_{11} & l_{21} & 0 & 0 \\ & l_{22} & l_{32} & 0 \\ 0 & & l_{33} & l_{43} \\ & & & l_{44} \end{pmatrix}$$

↳ fait le produit et déduire les l_{ii}

2) on a : dét $A_k \neq 0$, $\forall k = 1 \dots 4$

d'où A est décomposable en $A = L.U$

$A = L . U$

$$L = \begin{pmatrix} 1 & & & 0 \\ l_2 & 1 & & \\ 0 & l_3 & 1 & \\ 0 & 0 & l_4 & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} d_1 & c_1 & 0 & 0 \\ 0 & d_2 & c_2 & 0 \\ 0 & 0 & d_3 & c_3 \\ 0 & 0 & 0 & d_4 \end{pmatrix}$$

$c_1 = c_2 = c_3 = -1$

$d_1 = 2$

$l_2 d_1 = -1 \Rightarrow l_2 = -1/d_1$

$l_3 = \frac{-1}{d_2}$; $l_4 = \frac{-1}{d_3}$

$d_2 = 2 + l_2$

$d_3 = 2 + l_3$

$d_4 = 2 + l_4$

↳ par identification on peut déterminer les d_i, $i = 1 \dots 4$

Exercice 3

1) $\begin{cases} \Delta_n = \det A_n \\ \Delta_0 = 1 \end{cases}$, $\Delta_1 = \det A_1 = a_1$

$\begin{cases} \Delta_n = a_n \Delta_{n-1} - b_n C_{n-1} \Delta_{n-2} & n \geq 2 \\ \Delta_1 = a_1 \ ; \ \Delta_0 = 1 \end{cases}$

pour $n = 2$:

$\Delta_2 = a_2 a_1 - b_2 C_1 \Delta_0$

2) $A_n = LU$ ssi

$\Delta_i \neq 0$ $\forall i = 1 \dots n$

3)

$A_n = L.U$

$$L = \begin{pmatrix} 1 & & & & 0 \\ l_2 & 1 & & & \\ & l_3 & 1 & & \\ 0 & & \ddots & \ddots & \\ & & & l_n & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} d_1 & r_1 & & & 0 \\ 0 & d_2 & r_2 & & \\ & 0 & \ddots & \ddots & \\ & & & d_{n-1} & r_{n-1} \\ 0 & & & & d_n \end{pmatrix}$$

$d_1 = a_1$, $r_1 = c_1$

$\boxed{r_i = c_i, \ i = 1 \dots n-1}$

$l_2 d_1 = b_2 \rightarrow l_2 = \frac{b_2}{d_1}$

$l_3 d_2 = b_3 \Rightarrow l_3 = \dfrac{b_3}{d_2}$

$\hookrightarrow \boxed{l_i = \dfrac{b_i}{d_{i-1}} \quad , i = 2 \dots n}$

$l_2 r_1 + d_2 = a_2$

$\Rightarrow d_2 = a_2 - l_2 r_1$

$d_2 = a_2 - \dfrac{b_2}{d_1} r_1$

$\hookrightarrow \boxed{\begin{array}{l} d_i = a_i - \dfrac{b_i}{d_{i-1}} c_{i-1} \quad , i = 2 \dots n \\ d_1 = a_1 \end{array}}$

4)

$A_n . x = e$

$LUx = e \quad - \begin{cases} Ly = e & 1^{\text{a}} \\ Ux = y & 2^{\text{ème}} \end{cases}$

$e = \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$

$y_1 = e_1$

$l_2 y_1 + y_2 = e_2$

$y_2 = e_2 - l_2 y_1$

$\vdots$

$y_i = e_i - l_i y_{i-1}$

$(i = 2 \dots n)$

$\begin{cases} y_1 = e_1 \\ y_i = e_i - l_i y_{i-1} \ ; \ i = 2 \dots n \end{cases}$

$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

$x_n d_n = y_n \Rightarrow x_n = \dfrac{y_n}{d_n}$

$d_{n-1} * x_{n-1} + r_{n-1} x_n = y_{n-1}$

$\Rightarrow x_{n-1} = \left(y_{n-1} - r_{n-1} \dfrac{y_n}{d_n}\right) \dfrac{1}{d_{n-1}}$

$x_i = \left(y_i - r_i \dfrac{y_{i+1}}{d_{i+1}}\right) \dfrac{1}{d_i}$

$i = n-1 \dots 1.$

<u>Exercice 2</u>

1) α . tq $A = LL^T$

$A = \begin{pmatrix} 1 & \alpha & 0 \\ \alpha & 1 & \alpha \\ 0 & \alpha & 1 \end{pmatrix}$

$\begin{cases} 1 - \alpha^2 > 0 \\ 1 - \alpha^2 - \alpha^2 > 0 \Rightarrow 1 - 2\alpha^2 > 0 \end{cases}$

$\alpha \in \left] -\dfrac{\sqrt{2}}{2} ; \dfrac{\sqrt{2}}{2} \right[$

2) facile à faire.

<u>Exercice 4</u>

$$\begin{array}{cc} A_n = \begin{pmatrix} 2 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & -1 \\ 0 & & & -1 & 2 \end{pmatrix} & \begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{array} \end{array}$$

$\hookrightarrow$ matrice du Laplacien

$x_i = i x_1 , i = 1 \dots n$

$2x_1 - x_2 = 0 \Rightarrow x_2 = 2x_1$

$-x_1 + 2x_2 - x_3 = 0$

$\Rightarrow x_3 = -x_1 + 2x_2$

$\Rightarrow x_3 = -x_1 + 4x_1$

$x_3 = 3x_1$

$\vdots$

$-x_{n-2} + 2x_{n-1} - x_n = 0$

$\Rightarrow x_n = -x_{n-2} + 2x_{n-1}$

$x_n = -(n-2)x_1 + 2(n-1)x_1$

$x_n = n x_1$

$-x_{n-1} + x_n = 0$

$\Rightarrow 2x_n = x_{n-1}$

b) $2n x_1 = (n-1)x_1$

$\Rightarrow x_1 = 0$